PCET's Pimpri Chinchwad University

School of Computer Applications

MCA-II SEM-IV

Crypto and Blockchain

Unit No. 02 Cryptocurrency Regulations

Introduction

Cryptocurrency is legal to own and trade in India, where it is classified as "Virtual Digital Assets" (VDAs) rather than legal tender. Since 2022, a strict tax regime is in place (30% tax on gains, 1% TDS), and crypto exchanges must adhere to KYC and AML norms under the PMLA. The RBI continues to warn against risks, while exploring its own Central Bank Digital Currency (CBDC).

Key Aspects of Cryptocurrency Regulations in India (2026)

- **Legal Status:** Cryptocurrencies are not banned but are not recognized as legal tender. They are classified as Virtual Digital Assets (VDAs).
- **Taxation:** A 30% tax is levied on income from the transfer of VDAs, with no deduction for expenses other than the cost of acquisition
- **TDS (Tax Deducted at Source):** A 1% TDS applies to all VDA transactions, helping the government track trading volumes.
- **AML/KYC Compliance:** As of 2023, crypto exchanges and intermediaries are treated as reporting entities under the Prevention of Money Laundering Act (PMLA). They must perform KYC on users and report suspicious transactions to the Financial Intelligence Unit (FIU).
- **RBI Position:** The Reserve Bank of India maintains a cautious stance, highlighting risks regarding volatility and potential for money laundering. The RBI is focusing on developing its own digital currency, the CBDC.
- **No Formal Regulation Framework:** As of early 2026, there is no comprehensive law governing the licensing, operation, or consumer protection of crypto exchanges, although the PMLA classification acts as a form of regulation.
- **Future Outlook:** The government continues to monitor the space, balancing the need for innovation with financial stability risks.

Stakeholders

Cryptocurrency regulation stakeholders include government bodies, financial authorities, industry participants, and consumers aiming to balance innovation with risk mitigation. Key actors involve regulators (central banks, tax authorities), exchanges, investors, and legal bodies, with a focus on compliance, investor protection, and preventing financial crime.

Key Stakeholders in Cryptocurrency Regulation

- **Government & Regulatory Bodies (Internal & External):**
  - **Central Banks (e.g., RBI in India):** Formulate policies, monitor financial stability, and may issue Central Bank Digital Currencies (CBDCs).
  - **Financial Intelligence Units (FIU):** Act as watchdogs against money laundering and enforce compliance with anti-money laundering (AML) laws, requiring exchanges to report transactions.
  - **Tax Authorities (e.g., CBDT in India):** Focus on tax compliance, such as Tax Deducted at Source (TDS) on virtual digital asset (VDA) transactions.
  - **Securities Regulators (e.g., SEBI in India, SEC in the US):** Supervise market integrity and determine if assets qualify as securities.
  - **Law Enforcement Agencies:** Enforce rules against illicit activities and financial offenses.
- **Industry Stakeholders:**
  - **Cryptocurrency Exchanges (VASPs - Virtual Asset Service Providers):** Must comply with regulations, perform KYC (Know Your Customer), and implement reporting standards.
  - **Blockchain Developers & Miners:** Create and maintain the technical infrastructure.
  - **Financial Institutions:** Banks and traditional finance entities dealing with crypto-assets.
- **Users & Advocacy Groups:**
  - **Individual & Institutional Investors:** Seek protection and market stability.
  - **Merchants & Consumers:** Use crypto for payments, requiring secure frameworks.
  - **Industry Associations:** Lobby for regulations and standards.

Roots of Bitcoin

Bitcoin’s roots in blockchain began with Satoshi Nakamoto’s 2008 white paper, proposing a decentralized, peer-to-peer electronic cash system to bypass financial intermediaries. The blockchain, acting as a distributed ledger, was implemented on January 3, 2009, with the "genesis block" (block 0), which contained a headline about bank bailouts, marking its origin as a response to the 2008 financial crisis.

Key Aspects of Bitcoin's Blockchain Roots:

- **The Genesis Block:** Mined by Nakamoto on January 3, 2009, this first block established the foundation of the chain and holds 50 Bitcoin.
- **Cryptographic Foundation:** While Bitcoin was the first practical application, its roots lie in earlier, un-developed concepts like B-Money and Bit Gold.
- **Decentralized Ledger:** The blockchain ensures security and transparency by recording transactions across a network of computers, making it immune to central control or tampering.
- **Technical Structure:** Blocks are created roughly every 10 minutes, with each block referencing the hash of the previous one, forming an immutable chain.

Legal Aspects of Cryptocurrency Exchange

Legal aspects of cryptocurrency exchanges in 2026 center on strict Anti-Money Laundering (AML)/Know Your Customer (KYC) compliance, with many jurisdictions requiring registration as Virtual Asset Service Providers (VASPs). Exchanges must navigate evolving regulations regarding security, data protection, taxation on trading profits (e.g., a 30% tax in India), and user identity verification to combat illicit activities.

Key legal considerations for cryptocurrency exchanges include:

- **Regulatory Compliance & Licensing:** Exchanges are often required to register with financial intelligence units or regulatory bodies (e.g., FIU-IND in India) to operate legally.
- **AML and KYC Mandates:** Mandatory procedures to verify user identities, such as liveness detection, and monitoring transactions to prevent money laundering are required in most jurisdictions.
- **Taxation:** Income from trading or transferring virtual digital assets (VDAs) is subject to specific taxes, such as a 30% flat tax on gains and a 1% tax deducted at source (TDS) on transactions, as seen in India.

- **Legal Status of Assets:** While not legal tender, cryptocurrencies are treated as tradeable digital assets, necessitating clear terms of service regarding asset ownership and custody.
- **Security and Consumer Protection:** Regulations require robust cybersecurity measures to protect customer data and funds, with clear protocols for dispute resolution.
- **International Variations:** Legal definitions and regulations vary significantly, requiring compliance with the specific laws of each operating jurisdiction.

Black Market and Global Economy in Cryptocurrency Exchange

Cryptocurrency exchanges operate within the global economy while simultaneously facilitating black market activities through decentralized, often anonymous, transactions. Illicit actors use digital assets for money laundering, tax evasion, and purchasing prohibited goods, with roughly $76 billion of illegal activity annually involving Bitcoin

. These transactions often flow through mainstream exchanges to convert crypto to fiat currency.

Black Market Usage in Crypto Exchanges

- **Anonymity and Speed:** Cryptocurrencies provide a layer of anonymity, making it difficult to monitor illegal transactions.
- **Illicit Goods and Services:** Platforms have been used to purchase drugs, weapons, and to launder money, as seen in the infamous Silk Road case, which involved $1.2 billion in transactions.
- **Privacy Coins:** Digital assets like Monero (XMR), Zcash (ZEC), and Dash (DASH) are preferred for illegal activities due to enhanced anonymity.
- **Mixing Services:** Illicit actors use "mixers" to obscure the trail of funds, allowing them to deposit laundered assets into mainstream exchanges.

Impact on the Global Economy

- **Significant Capital Flow:** While often used for illicit purposes, the overall crypto market, with a market capitalization reaching approximately $2.98 trillion by late 2025, has become a major, albeit volatile, part of the financial landscape.
- **Regulatory Concerns:** Governments are increasingly concerned about the misuse of digital currencies for terrorist financing, tax evasion, and money laundering.
- **Market Manipulation:** Research suggests potential for significant market manipulation, affecting the stability and trust in these exchanges.

- **Alternative Asset:** Despite risks, cryptocurrencies are viewed by some as a hedge against traditional market fluctuations, especially during times of crisis.

Cryptocurrency Applications Internet of Things

The tension between the need for regulation and the decentralized nature of cryptocurrencies continues to define their role in the global economy, necessitating advanced methods to mitigate cybercriminality.

Cryptocurrency and blockchain technology enhance Internet of Things (IoT) applications by providing secure, decentralized, and automated machine-to-machine (M2M) transactions. Key use cases include automated payments for data/resources, improved security, smart contract automation, supply chain tracking, and data integrity. Common applications, such as IOTA, enable feeless microtransactions, while blockchain reduces reliance on central intermediaries for IoT management.

Key Cryptocurrency & Blockchain Applications in IoT

- **Machine-to-Machine (M2M) Transactions:** Devices, such as smart meters or autonomous vehicles, can buy or sell services like electricity, bandwidth, or data without human intervention, often using specialized cryptocurrencies like IOTA.
- **Secure Data Sharing & Integrity:** Blockchain creates immutable, tamper-resistant records of data generated by IoT devices.
- **Smart Contract Automation:** Automated agreements and payments can be executed based on IoT data triggers, such as releasing payment upon delivery confirmation in a supply chain.
- **Decentralized IoT Management:** Rather than relying on a central server prone to failure, blockchain allows for decentralized control and authentication of IoT devices.
- **Supply Chain & Logistics:** Real-time tracking with IoT sensors paired with blockchain ensures secure, auditable, and transparent, end-to-end monitoring.

Platforms and Technologies

- **IOTA (Tangle):** Specifically designed for IoT to enable high-speed, feeless, microtransactions for M2M interactions.
- **Ethereum & Hyperledger:** Used for implementing smart contracts and permissioned networks to manage IoT device interactions and security.

Benefits to IoT

- **Enhanced Security:** Combines asymmetric cryptography with decentralized ledgers to protect against data tampering and unauthorized access.
- **Reduced Costs:** Eliminates middlemen in transactions.
- **Scalability & Trust:** Enables trusted interactions among unauthorized IoT devices.

Cryptocurrency Applications Medical Record Management System

Cryptocurrency and blockchain technologies are revolutionizing medical record management by providing decentralized, secure, and immutable storage solutions. Using blockchain (e.g., Ethereum, Hyperledger), patient records are encrypted and stored across distributed nodes, ensuring data integrity, privacy, and HIPAA compliance. Smart contracts automate access, allowing patients to control and share data with providers via secure wallets, enhancing interoperability.

Key Applications of Blockchain in Medical Records

- **Immutable Data Storage:** Medical records are stored as immutable, time-stamped, encrypted records, creating a secure "digital fingerprint" that prevents tampering or unauthorized changes.
- **Decentralized Access Management:** Smart contracts enable patient-controlled access, allowing patients to manage, share, or revoke access to their health records using cryptographic keys.
- **Patientory & Health Wallets:** Applications like Patientory function as global, secure, blockchain-based wallets, allowing users to manage, share, and track their personal health information.
- **Secure Data Exchange:** Blockchain provides a secure, interoperable platform for sharing records between providers, which is essential for, for example, cross border or cross-institutional patient care.
- **Consent Management:** Blockchain can record and track patient consent permissions for sharing data among medical institutions.
- **Audit Trails:** Every interaction with a patient's record is logged on the blockchain, providing a transparent, unalterable trail for auditing purposes.

Benefits of Blockchain in Healthcare Records

- **Security & Data Integrity:** Research from the NIH highlights that decentralized networks eliminate single points of failure, protecting against data breaches.

- **Reduced Costs & Efficiency:** Smart contracts remove the need for manual record management, reducing administrative costs and speeding up data retrieval for treatment.
- **Interoperability:** Blockchain can act as a standardized, secure layer that allows disparate, independent systems to exchange patient information seamlessly.

Examples in Practice

- MIT Media Lab's MedRec**:** A pilot program using blockchain for decentralized content access management.
- **HyperLedger Fabric/Indy:** Used to build EHR systems (e.g., "Que Bian") that manage identities for patients, doctors, and hospitals.
- IPFS-based Systems**:** Using the InterPlanetary File System (IPFS) in conjunction with blockchain to store large encrypted records off-chain, while anchoring hash references on the chain for security.

Domain Name System and Future of Blockchain

Blockchain DNS revolutionizes the traditional Domain Name System (DNS) by shifting from centralized, registrar-based management to decentralized, user-owned, and censorship-resistant, blockchain-based infrastructure. By storing domain records on public ledgers, these systems enhance security against hijacking, allow permanent ownership, and directly integrate with crypto wallets to replace complex addresses with human-readable names like. eth or. crypto.

Key Aspects of Blockchain DNS and the Future:

- **Decentralization & Security:** Unlike the traditional DNS, which relies on the ICANN and centralized registries, blockchain domains (e.g., Unstoppable Domains, ENS) are managed via decentralized smart contracts, removing single points of failure.
- **True Ownership & No Fees:** Users hold private keys to their domains, providing permanent ownership without renewal fees, whereas traditional domains are rented through registrars.
- **Web3 Integration:** Blockchain domains function as digital identity passports, enabling secure website navigation, cryptocurrency payments, and decentralized app (dApp) authentication.
- **Future Trends:** Future development aims at enhancing interoperability with Web2, improving privacy through Zero-Knowledge Proofs, and resisting quantum computing threats.

- **Hybrid Future:** The future likely involves a hybrid approach, where traditional DNS and decentralized naming systems coexist, allowing organizations to balance security, compliance, and user-centric control.

Decentralization and Cryptography

Decentralization in cryptography refers to distributing control and data across a network of nodes, eliminating single points of failure, central authorities, or intermediaries. Cryptographic techniques, such as public-key cryptography and digital signatures, secure these systems, enabling trustless, peer-to-peer, and tamper-resistant transactions and data storage.

Key Aspects of Decentralization and Cryptography

- **Security Through Consensus:** Cryptographic algorithms are used to verify transactions, which are then recorded on a decentralized, distributed ledger (blockchain).
- **Cryptographic Tools:** Asymmetric cryptography (public/private keys) is fundamental, allowing secure, direct interactions without a trusted third party.
- **No Single Point of Failure:** By removing central, trusted authorities (like banks or corporations), decentralized networks become more resilient to attacks and corruption.
- **Trustless Environment:** Users don't need to trust each other or a central authority; they trust the underlying mathematics and code.
- **Applications:** Beyond cryptocurrencies, this combination is used for decentralized finance (DeFi), secure messaging, data storage, and self-sovereign identity.

Key Concepts in Decentralized Cryptography

- **Nodes:** The distributed, independent computers that participate in the network, maintaining the ledger and verifying transactions.
- **Consensus Mechanism:** Protocols, such as Proof-of-Work or Proof-of-Stake, that enable nodes to reach an agreement on the state of the ledger without a central, trusted party.
- **Smart Contracts:** Self-executing, automated, and tamper-proof contracts on a blockchain, which eliminate the need for traditional, centralized enforcement.
- **Distributed Storage:** Storing data across a network, ensuring that information is not controlled by a single entity.

Benefits and Challenges

- **Benefits:** Increased privacy, censorship resistance, improved security, and enhanced accessibility.

- **Challenges:** Scalability issues, high energy consumption (for some proof-of-work systems), regulatory uncertainty, and potential for "hidden" centralization in practice.
- **Spectrum of Decentralization:** Decentralization is not binary; it's a spectrum, with some networks being more decentralized than others based on node distribution, governance, and development.